

Python OOP Complete Guide

This PDF covers core Object Oriented Programming concepts in Python including classes, objects, constructor (`__init__`), `self`, arguments, operator precedence, inheritance, polymorphism, abstraction and encapsulation with examples.

1. Class and Object

Class ek blueprint hota hai aur object us class ka instance hota hai. Example: `class Car: pass`
`car1 = Car()`

2. `__init__` Constructor

`__init__` automatically call hota hai jab object banaya jata hai. Example: `class Car: def __init__(self, name): self.name = name`

3. `self` keyword

`self` current object ko represent karta hai. Is ke through hum attributes access karte hain.

4. Arguments

Functions aur methods ko data pass karne ke liye arguments use hotay hain.

5. Operator Precedence

Python me operations ki priority hoti hai: BODMAS rule follow hota hai. Example: $2 + 3 * 4 = 14$

6. Inheritance

Ek class doosri class ki properties inherit karti hai. Example: `class Parent: pass`
`class Child(Parent): pass`

7. Polymorphism

Same function different behavior show karta hai. Example: `len()` string aur list dono par work karta hai

8. Encapsulation

Data ko hide karna aur private banana. Example: `__variable` private hota hai

9. Abstraction

Complex implementation hide karna aur sirf essential features show karna.